

Notes: Kleven, Knudsen, Kreiner, Pedersen, and Saez (Econometrica, 2011)

Unwilling or Unable to Cheat? Evidence from a Tax Audit Experiment in Denmark

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1 Big picture

- **Question.** Why is tax compliance high in modern economies despite low audit probabilities and modest penalties? Are taxpayers mostly unwilling to cheat, or are they unable to cheat because income is third-party reported?
- **Core distinction.** The paper separates income subject to third-party reporting from self-reported income. This distinction is the central organizing variable in the model, experiment, and interpretation.
- **Empirical setting.** The authors collaborate with the Danish tax authority on a large randomized tax audit and audit-threat field experiment involving more than 40,000 individual income tax filers.
- **Main baseline audit result.** Evasion is close to zero for income subject to third-party reporting but substantial for self-reported income. Since most Danish income is third-party reported, the aggregate evasion rate is modest.
- **Main tax-rate result.** Marginal tax rates affect evasion for self-reported income, but the evasion response is small relative to legal avoidance and other behavioral responses.
- **Main enforcement result.** Prior audits and threat-of-audit letters increase later reported income for self-reported components, but have essentially no effect on third-party-reported income.
- **Model contribution.** The paper extends the Allingham-Sandmo model by allowing the detection probability to depend on the type and amount of evasion. Third-party reporting makes detection almost automatic once evasion crosses the self-reported margin.
- **Policy takeaway.** Information reporting and withholding are more important than low audit probabilities or small penalties in explaining high compliance in advanced tax systems.
- **Conceptual takeaway.** The answer is both: many taxpayers may be unwilling to cheat, but the central institutional reason for high compliance is that most income is hard to conceal.

2 Incremental contribution of the paper

2.1 Contribution relative to the Allingham-Sandmo model

- The classic model treats detection probability as exogenous and usually independent of the form of income being evaded.
- Kleven et al. make the detection technology endogenous to the information structure of the tax base.
- Their model explains why low audit probabilities can coexist with high compliance: third-party reporting makes the effective detection probability high for most income even when formal audit rates are low.

Interpretation: The key incremental step is to model tax evasion not as a single homogeneous decision but as a portfolio of income items with different enforcement technologies.

2.2 Contribution relative to audit-based evasion estimates

- Audit studies usually measure evasion after the fact, but often do not combine audit data with randomized enforcement treatments.
- This paper has a baseline audit that measures evasion and then a second-stage experiment that manipulates future audit probabilities.
- The same taxpayers are observed under randomized audit/no-audit histories and randomized letter treatments.

Interpretation: The paper does not merely estimate the amount of evasion. It experimentally identifies how taxpayers respond to enforcement.

2.3 Contribution relative to taxable-income elasticity research

- A large literature uses bunching around tax schedule kinks to estimate taxable income responses.
- This paper uses administrative audit data to decompose the bunching response into evasion versus legal avoidance or real behavioral responses.
- The finding is that evasion contributes only a modest part of the bunching response for self-reported income and almost none for third-party-reported income.

Interpretation: This connects two literatures that often remain separate: bunching around tax kinks and direct audit evidence on noncompliance.

2.4 Contribution relative to tax enforcement field experiments

- The field experiment is large, nationally representative within strata, and run by the tax authority in a real enforcement environment.
- The design separates actual prior audits from announced future audit threats.
- It also separates the effects on third-party-reported income and self-reported income.

Interpretation: The paper provides unusually clean evidence on where enforcement works: it works mainly where taxpayers have reporting discretion.

3 Data and experimental design

3.1 Danish tax system and reporting structure

- Denmark is a high-tax, high-information-reporting economy.
- Most labor income, transfer income, interest income, pension income, and many deductions are subject to third-party reporting.
- Tax returns are pre-populated with third-party information, and taxpayers can amend or add information when filing.
- Self-reported income includes categories such as self-employment income and some capital income items, where taxpayers have more discretion.

Interpretation: The Danish environment is ideal for separating income that is essentially visible to the tax authority from income that is more privately reported.

3.2 Baseline sample

- The experiment uses a stratified sample of more than 40,000 individual filers.
- The sample includes employees and self-employed individuals.
- Stratification ensures representation across tax return complexity and administrative regions.
- The baseline year is income earned in 2006 and filed in 2007.

Interpretation: Employees mostly have third-party-reported income, while self-employed individuals have more self-reported income. This creates a natural contrast in evasion opportunities.

3.3 First-stage audit randomization

Taxpayers are randomly assigned to:

- a 100% audit group, whose 2006 returns are thoroughly audited;
- a 0% audit group, whose returns are deliberately not audited.

The audits are comprehensive and examine every tax item using standard verification procedures. Audit adjustments can be positive or negative.

Interpretation: The first-stage randomization produces causal variation in prior audit experience and also creates high-quality measurement of baseline evasion.

3.4 Second-stage letter randomization

In the following year, employees in both first-stage groups are randomly assigned to:

- no letter;
- a letter announcing a 50% probability of audit;
- a letter announcing a 100% probability of audit.

The 2007 returns filed in 2008 are then audited according to the announced probabilities.

Interpretation: The letter experiment tests deterrence and responsiveness to audit threats, separately from the effect of having been audited in the previous year.

4 Model and theoretical predictions

4.1 Taxpayer problem

Let y be true income, e be evaded income, τ be the tax rate, and θ be the penalty rate. A simple expected utility representation is:

$$(1 - p(e))u(y - \tau(y - e)) + p(e)u(y - \tau y - \theta\tau e), \quad (1)$$

where $p(e)$ is the probability that evasion is detected.

Interpretation: The key departure from the simplest model is that $p(e)$ is not constant. It is low for evasion of self-reported income and rises sharply once evasion extends into third-party-reported income.

4.2 Detection probability under third-party reporting

The model assumes a detection probability shaped like Figure 1:

- $p(e)$ is initially close to zero for small amounts of evasion within self-reported items;
- $p(e)$ rises steeply as the taxpayer begins to understate items known to third parties;
- $p(e)$ approaches one once evasion covers income that is automatically reported by employers, banks, funds, or other institutions.

Interpretation: The S-shaped detection technology generates a strong prediction: evasion should concentrate where income is self-reported and should be very low where income is third-party reported.

4.3 Predictions

The model implies:

1. Evasion is substantial for self-reported income.
2. Evasion is close to zero for third-party-reported income.
3. Marginal tax rates increase evasion mainly for self-reported income.
4. Audits and audit threats affect self-reported income but have little effect on third-party-reported income.
5. Tax enforcement effects should be largest among taxpayers with more self-reported income.

Interpretation: The empirical design directly tests each of these predictions.

5 Empirical specifications

5.1 Audit adjustment decomposition

For audited taxpayer i , the audit adjustment for income component k is

$$Adj_{ik} = AuditedIncome_{ik} - PreAuditIncome_{ik}. \quad (2)$$

The adjustment is decomposed into:

$$Adj_{ik} = Underreporting_{ik} - Overreporting_{ik}. \quad (3)$$

Interpretation: Positive adjustments imply underreported tax bases. Negative adjustments imply overreported income or deductions.

5.2 Probability of underreporting

The paper estimates linear probability models of the form:

$$1\{Adj_{ik} > 0\} = \alpha + X'_i\beta + \varepsilon_i, \quad (4)$$

where X_i includes social factors, socioeconomic variables, tax return factors, and self-reported income indicators.

Interpretation: The strongest predictor of underreporting is the presence and amount of self-reported income.

5.3 Bunching and marginal tax rates

The paper uses large kinks in the Danish income tax schedule to estimate bunching. Excess mass around a kink is translated into an elasticity. Let B be bunching mass and $h_0(z)$ be the counterfactual density. A stylized elasticity is:

$$e = \frac{\Delta z / z}{\Delta(1 - \tau) / (1 - \tau)}. \quad (5)$$

The paper compares elasticities before and after audit adjustment to separate legal avoidance plus real responses from evasion.

Interpretation: If bunching disappears after audit adjustment, the bunching was evasion. If bunching remains, the response is legal avoidance or real behavior.

5.4 Prior audit and threat-letter effects

The randomized enforcement effects are estimated with regressions such as:

$$\Delta Income_i = \alpha + \beta Audit_i + \gamma Letter50_i + \delta Letter100_i + X_i' \eta + \varepsilon_i. \quad (6)$$

The paper estimates effects on total reported income, self-reported income, third-party-reported income, and audit adjustments.

Interpretation: Random assignment identifies causal deterrence effects. Effects should appear only where taxpayers have discretion.

6 Identification and empirical credibility

6.1 Randomized audits and letters

- The baseline audit and no-audit groups are randomly assigned.
- Threat letters are randomly assigned in the following year.
- Randomization checks show the groups are balanced on baseline income, deductions, demographic characteristics, and return complexity.

Interpretation: The randomized design separates enforcement effects from taxpayer selection and administrative targeting.

6.2 Audit data and measurement

- Unlike survey-based compliance measures, the audit data measure actual corrections to tax returns.
- The audits are comprehensive and cover all tax items.
- The distinction between third-party and self-reported income is observed directly in administrative data.

Interpretation: The paper has unusually strong measurement of both evasion and the information environment.

6.3 Why the bunching design adds value

- The bunching analysis uses tax schedule kinks to test behavioral responses to tax rates.
- Audit data allow the authors to observe how much of the pre-audit bunching remains after evasion is corrected.
- This helps distinguish evasion from legal tax avoidance or real labor-supply responses.

Interpretation: Combining bunching and audits is a major strength: it lets the paper decompose taxable income elasticities.

6.4 Limitations

- Denmark has unusually strong third-party reporting and high administrative capacity, so external validity depends on information-reporting institutions.
- The field experiment focuses on individual taxpayers, not corporate tax evasion.

- Audits reveal detected evasion; some sophisticated forms of evasion may remain unobserved even under audit.
- Threat letters are studied in a particular institutional context with high trust in tax administration.

Interpretation: The findings are strongest for modern personal income tax systems with extensive information reporting.

7 Tables and figures

The figures and tables below are directly cropped from the paper. Adjacent figures and tables are placed on the same row whenever possible; a single item is displayed at width $0.6 \backslash \text{linewidth}$.

7.1 Figure 1 and Table I: model intuition and Danish tax system

Read:

- Figure 1 illustrates the detection probability under third-party reporting.
- Detection is low for evasion within self-reported items but rises rapidly once evasion reaches third-party-reported income.
- Table I defines Danish income concepts and tax rates in 2006.
- Labor income, personal income, capital income, deductions, taxable income, and stock income are different tax bases.

Economic message: Third-party reporting changes the economics of evasion by making large parts of the income base difficult to hide.

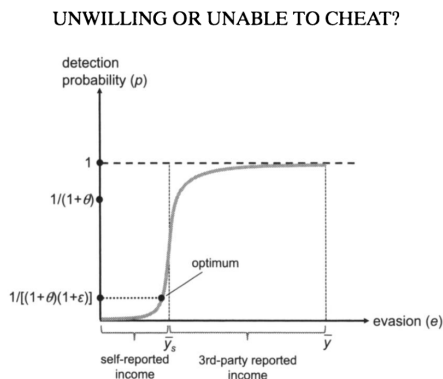


FIGURE 1.—Probability of detection under third-party reporting.

of tax returns and information reports will recover any evasion
Figure 1: detection probability under third-party reporting

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TABLE I
DANISH INDIVIDUAL INCOME TAX IN 2006

A. Income Concepts			
Income Concept	Definition		
1. Labor income	Salary, wages, honoraria, fees, bonuses, fringe benefits, business earnings		
2. Personal income	Labor income (1) + social transfers, grants, awards, gifts, received alimony – payroll tax, and certain pension contributions		
3. Capital income	Interest income, rental income, business capital income – interest on debt (mortgage, bank loans, credit cards, student loans)		
4. Deductions	Commuting costs, union fees, unemployment contributions, other work related expenditures, charitable contributions, alimony paid		
5. Taxable income	= Personal income (2) + capital income (3) – deductions (4)		
6. Stock income	Dividends and realized capital gains from corporate stock		
B. Tax Rates and Tax Bases			
Tax Type ^a	Tax Base	Bracket (DKK) ^b	Tax Rate
Payroll tax	Labor income	All income	8.0%
		38,500–265,500	5.5%
National income tax	Personal income + max(capital income, 0)	265,500–318,700	11.5%
		318,700–	26.5% ^c
Regional income tax	Taxable income	38,500–	32.6% ^d
Stock income tax	Stock income	0–44,400	28.0%
		44,400–	43.0%

^aThe national and regional income taxes are based on individual income (not family income). The stock income tax is based on family income with brackets for married tax filers twice as large as those reported in the table.

^bAll amounts are given in Danish kroner: U.S. \$1 = 5.2 DKK as of January 2010.

^cThe top rate is reduced so that the combined national and regional income top marginal tax rate never exceeds 59%. The top marginal tax rate on labor income including the payroll tax is therefore $0.08 + 0.59 = 62.3\%$.

^dThe regional tax includes municipal and county taxes in 2006. The rate shown is the average across all municipalities, and includes the optional church tax equal to 0.7%.

Table I: Danish individual income tax in 2006

7.2 Figure 2 and Table II: experimental design and audit adjustments

Read:

- Figure 2 shows the two-stage randomized audit and letter design.
- Table II decomposes audit adjustments by income type and reporting status.
- Underreporting is concentrated in self-reported income.
- Third-party-reported income shows very small underreporting relative to its size.

Economic message: The audit data directly reveal that evasion opportunities are concentrated where information reporting is weak.

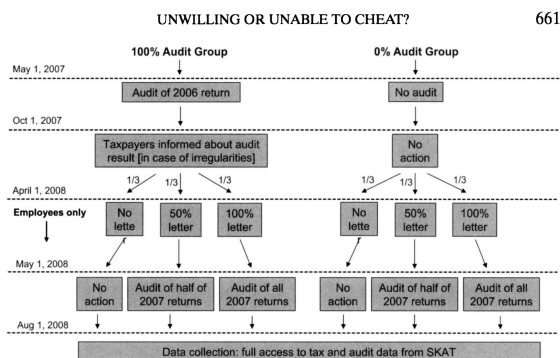


Figure 2: overview of experiment

TABLE II—Continued

	A. Total Income Reported				B. Third-Party vs. Self-Reported Income			
	Pre-Audit Income	Audit Adjustment	Under-reporting	Over-reporting	Third-Party Income	Third-Party Under-reporting	Self-Reported Income	Self-Reported Under-reporting
	1	2	3	4	5	6	7	8
III. Income Components								
Personal income								
Amounts	210,178	2327	2398	-71	211,244	463	-1066	1936
	(1481)	(399)	(399)	(11)	(1385)	(74)	(548)	(392)
% Nonzero	95.22	2.49	1.99	0.50	95.20	1.30	11.95	0.82
	(0.15)	(0.11)	(0.10)	(0.05)	(0.15)	(0.08)	(0.23)	(0.06)
Capital income								
Amounts	-11,075	254	286	-32	-14,556	98	3481	188
	(340)	(49)	(49)	(6)	(602)	(11)	(542)	(47)
% Nonzero	93.95	2.10	1.69	0.41	94.91	0.79	12.29	1.28
	(0.17)	(0.10)	(0.09)	(0.05)	(0.16)	(0.06)	(0.23)	(0.08)
Deductions								
Amounts	-9098	148	197	-49	-5666	18	-3432	179
	(104)	(17)	(15)	(7)	(48)	(3)	(85)	(15)
% Nonzero	60.07	3.45	2.56	0.89	57.61	0.31	22.60	2.49
	(0.35)	(0.13)	(0.11)	(0.07)	(0.35)	(0.04)	(0.30)	(0.11)

(Continues)

Table II, part 2: income components

TABLE II
AUDIT ADJUSTMENTS DECOMPOSITION*

	A. Total Income Reported				B. Third-Party vs. Self-Reported Income			
	Pre-Audit Income	Audit Adjustment	Under-reporting	Over-reporting	Third-Party Income	Third-Party Under-reporting	Self-Reported Income	Self-Reported Under-reporting
	1	2	3	4	5	6	7	8
I. Net Income and Total Tax								
Net income	206,038	4532	4796	-264	195,969	612	10,069	4183
	(2159)	(494)	(493)	(31)	(1798)	(77)	(1380)	(486)
% Nonzero	98.38	10.74	8.58	2.16	98.57	2.31	38.18	7.29
	(0.09)	(0.22)	(0.20)	(0.10)	(0.08)	(0.11)	(0.35)	(0.19)
Total tax	69,940	1980	2071	-91				
	(1142)	(236)	(235)	(11)				
% Nonzero	90.76	10.59	8.41	2.18				
	(0.21)	(0.22)	(0.20)	(0.10)				
II. Positive and Negative Income								
Positive income	243,984	3776	3943	-167	223,882	516	20,102	3427
	(2511)	(485)	(485)	(27)	(1860)	(76)	(1693)	(478)
% Nonzero	98.24	5.80	4.78	1.02	98.15	1.60	19.53	3.41
	(0.09)	(0.17)	(0.15)	(0.07)	(0.10)	(0.09)	(0.28)	(0.13)
Negative income	-37,946	756	853	-97	-27,913	97	-10,033	756
	(1014)	(71)	(69)	(14)	(406)	(12)	(862)	(68)
% Nonzero	79.09	6.45	5.13	1.32	78.21	0.75	29.49	4.99
	(0.29)	(0.18)	(0.16)	(0.08)	(0.29)	(0.06)	(0.33)	(0.16)

(Continues)

Table II, part 1: audit adjustments

TABLE II—Continued

	A. Total Income Reported				B. Third-Party vs. Self-Reported Income			
	Pre-Audit Income	Audit Adjustment	Under-reporting	Over-reporting	Third-Party Income	Third-Party Under-reporting	Self-Reported Income	Self-Reported Under-reporting
	1	2	3	4	5	6	7	8
III. Income Components (Continued)								
Stock income								
Amounts	5635	259	281	-22	3783	30	1852	251
	(1405)	(45)	(45)	(8)	(976)	(12)	(943)	(43)
% Nonzero	22.47	0.95	0.80	0.15	22.44	0.07	2.45	0.75
	(0.30)	(0.07)	(0.06)	(0.03)	(0.30)	(0.02)	(0.11)	(0.06)
Self-employment								
Amounts	10,398	1544	1633	-89	1164	4	9234	1630
	(812)	(280)	(279)	(26)	(177)	(2)	(816)	(279)
% Nonzero	7.63	3.43	3.02	0.41	1.40	0.04	7.66	3.00
	(0.19)	(0.13)	(0.12)	(0.05)	(0.08)	(0.01)	(0.19)	(0.12)

* All amounts are in Danish kroner (1 US \$1 = 5.2 DKK as of 1/2010) and negative amounts (such as deductions) are reported in negative. Column 1 reports pre-audit amounts and the percent of files with nonzero pre-audit amounts. Column 2 displays the net audit adjustment (and percent with nonzero net audit adjustment); column 3 displays underreporting in the audit adjustment defined as downward audit adjustments increasing tax liability (and percent with underreporting); column 4 displays overreporting in the audit adjustment defined as downward audit adjustments decreasing tax liability (and percent with overreporting). Note that columns 3 + 4 = 2. Column 5 displays third-party income (and percent with nonzero third-party income); column 6 displays third-party income underreporting defined as upward audit adjustments in the case where third-party income is higher than final reported income for positive income items (and percent with third-party income underreporting); column 7 displays self-reported income defined as total reported income minus third-party reported income (and percent with nonzero self-reported income); column 8 displays self-reported income underreporting defined as all upward audit adjustments net of third-party income under-reporting (and percent with self-reported income under-reporting). Note that 5 + 7 = 1 and 6 + 8 = 3. Panel I reports net income (sum of all positive income components minus all negative income components and other deductions) and total tax. Panel II reports positive income (sum of all positive income components) and negative income (sum of all negative income components and deductions). Panel III displays various income components. Personal income is earnings, pensions, and alimony minus some retirement contributions. Capital income is interest income, returns on bonds, and net rents minus all interest payments. Deductions include work-related expenses, union fees, charitable contributions, alimony paid, and various smaller items. Stock income includes dividends and realized capital gains on stocks. Self-employment income is net profits from unincorporated businesses. Net income is personal income, and capital income, stock income, and self-employment income minus deductions. All estimates are population weighted and based solely on the 100% audit group (19,680 observations). Standard errors are reported in parentheses.

Table II, part 3: stock/self-employment continuation

7.3 Figure 3 and Table III: anatomy and predictors of evasion

Read:

- Figure 3 shows that, among evaders, the most common outcome is evasion of all self-reported income.
- The evasion rate increases with the fraction of income that is self-reported.
- Table III shows that self-reported income variables strongly predict underreporting.
- Social and socioeconomic variables explain less than the direct opportunity to misreport.

Economic message: Evasion is primarily about opportunity created by the reporting technology, not only about taxpayer preferences.

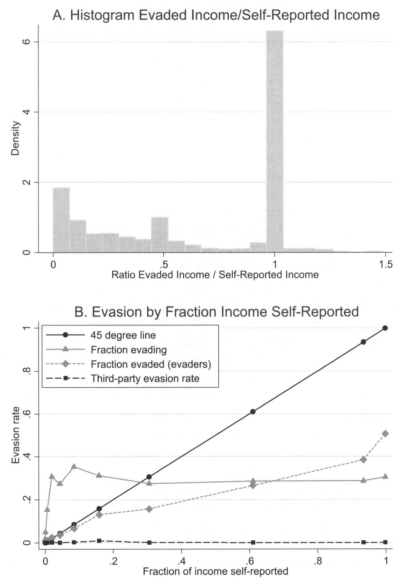


FIGURE 3.—Anatomy of tax evasion. Panel A displays the density of the ratio of evaded income to self-reported income (after audit adjustment) among those with a positive tax evasion of the 100% audit group and population weights. Income is defined as the sum of all positive items (so that self-reported income is always positive). Panel A shows that, among evaders, the most common act is to evade all self-reported income. About 70% of taxpayers with positive reported income do not have any adjustment and are not represented on panel A. Panel B displays the fraction of third-party income evaded (unconditional on evading) by deciles of fraction of income self-reported (after audit adjustment and adding as one category those with zero reported income). Panel B also displays the fraction of third-party income evaded (unconditional on evading) by deciles of fraction of income self-reported (after audit adjustment and adding as one category those with zero reported income).

Figure 3: anatomy of tax evasion

TABLE III
PROBABILITY OF UNDERREPORTING: SOCIOECONOMICS VERSUS TAX RETURN FACTORS*

Coefficients (in Percent)	A. Basic Variables				B. Detailed Variables			
	Social Factors	Socio-economic Factors	Tax Return Factors	All Factors	Social Factors	Socio-economic Factors	Tax Return Factors	All Factors
Constant	12.72 (1.06)	10.13 (1.12)	1.18 (0.25)	3.72 (1.01)	6.95 (1.64)	5.55 (2.16)	0.95 (2.04)	2.24 (2.99)
Female dummy	-5.56 (0.63)	-4.17 (0.65)	-2.06 (0.62)	-2.06 (0.62)	-5.29 (0.62)	-3.33 (0.67)	-1.02 (0.62)	-1.02 (0.62)
Married dummy	1.22 (0.70)	-0.55 (0.72)	-1.50 (0.72)	-1.50 (0.72)	-0.72 (0.77)	-1.98 (0.78)	-1.70 (0.78)	-1.70 (0.78)
Member of church	-1.59 (0.98)	-2.27 (0.97)	-0.94 (0.92)	-0.94 (0.92)	-1.54 (1.02)	-1.88 (0.99)	-0.71 (0.92)	-0.71 (0.92)
Geographical location	Copenhagen dummy (1.52)	-0.01 (1.51)	-0.25 (1.47)	-0.25 (1.47)	6 location dummies (1.47)	6.86 (8.87)	0.95 (8.87)	2.24 (33.53)
Age	Age > 45 dummy (0.67)	-0.72 (0.67)	-0.63 (0.61)	-0.56 (0.61)	4 age group dummies (0.61)	0.00 (0.73)	0.00 (0.73)	24.33 (0.71)
Home ownership	Firm size < 10 dummy (1.26)	5.49 (1.26)	0.15 (1.05)	0.15 (1.05)	5 firm size dummies (1.05)	3.72 (0.90)	0.00 (0.90)	-0.88 (0.90)
Industrial sector	Informal sector dummy (1.15)	0.27 (1.15)	0.27 (0.92)	0.27 (0.92)	22 industry dummies (0.92)	0.00 (0.73)	0.00 (0.73)	0.00 (0.71)
Self-reported income dummy			5.58 (0.75)	5.59 (0.80)			3.49 (0.80)	3.75 (0.78)
Self-reported income > 20,000 DKK			21.68 (1.38)	21.09 (1.40)			9.79 (1.62)	8.76 (1.61)
Self-reported income < -10,000 DKK			14.99 (1.42)	14.74 (1.42)			14.56 (1.41)	14.24 (1.38)

(Continued)

Table III, part 1: predictors of underreporting

TABLE III—Continued

Coefficients (in Percent)	A. Basic Variables				B. Detailed Variables			
	Social Factors	Socio-economic Factors	Tax Return Factors	All Factors	Social Factors	Socio-economic Factors	Tax Return Factors	All Factors
	1	2	3	4	5	6	7	8
Auditing flag dummy			13.22 (1.58)	13.07 (1.53)			12.26 (1.61)	12.37 (1.56)
Self-employed dummy							17.03 (1.14)	13.47 (1.39)
Capital income dummy							-0.75 (1.98)	-0.47 (1.87)
Stock income dummy							0.33 (0.65)	1.21 (0.66)
Deduction dummy							-1.12 (0.72)	-0.76 (0.88)
Audit adjustment in 2004 or 2005 dummy							7.22 (1.58)	6.86 (1.55)
Income controls							0.20 (0.20)	0.02 (0.02)
R-squares	1.16%	2.46%	16.15%	16.53%	2.16%	7.76%	18.72%	19.76%
Adjusted R-squares	1.14%	2.42%	16.14%	16.48%	2.11%	7.58%	18.66%	19.54%

*This table reports coefficients of the OLS regression of dummy for underreporting on various dummy regressors. All coefficients are expressed in percent and robust standard errors are reported. Bottom rows report the R-square and adjusted R-squares. All estimates are population weighted and based solely on the 100% audit group (19,680 observations). Standard errors reported in parentheses. In part A (columns 1–4), we include a basic set of dummy variables, while a richer set of variables is included in part B (columns 5–8). In part B, we do not report the full set of coefficients for geographical, age, firm size, industrial sector, and income groups. We instead only report the p-value from an F-test that the coefficients of those dummies are all equal to zero (for each category). The six location dummies are defined as Copenhagen, North Sealand, Middle and south Sealand, South Denmark, Middle Jutland, and North Jutland. The four age dummies are for age groups 0–25, 26–45, 46–65, and 66+. The five firm size dummies are for firms' size: 1, 2–10, 11–100, 101–1000, and 1001+. The six income group dummies are for each of the bottom three quartiles separately, percentile 75–95, percentile 95–99, and top percentile. For income categories, self-employed dummy means nonzero self-employment income and so forth.

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Table III, part 2: predictors of underreporting continued

7.4 Figure 4 and Table IV: tax rates, bunching, and evasion versus avoidance

Read:

- Figure 4 shows bunching around large tax kinks for self-employment income and stock income.
- Bunching remains even after audit adjustments, implying that much of it is legal avoidance or real response rather than illegal evasion.
- Table IV quantifies evasion versus avoidance elasticities.

- For self-employment income, the evasion elasticity is much smaller than the total pre-audit taxable-income elasticity.

Economic message: Marginal tax rates affect evasion, but legal avoidance and other responses account for most observed bunching around tax kinks.

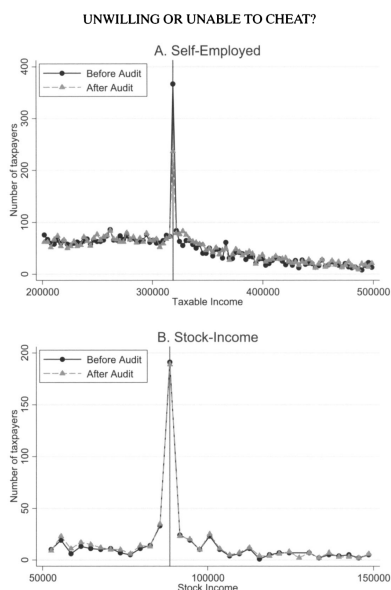


FIGURE 4.—Density distributions around kink points. The figure displays number of taxpayers by 3000 DKK bins for taxable income for the self-employed (panel A) and stock income (panel B). In both panels, we report the series for incomes before audits and incomes after audit for the 100% audit group. The vertical line denotes the kink point where marginal tax rates jump. The jump is from 49% to 62% in panel A (top taxable income bracket) and from 28% to 43% in panel B (top stock income bracket). For married filers, the stock income tax is assessed jointly in the bracket threshold in the figure is the one that applies to such joint filers. For single filers, the bracket threshold is half as large at 44,300 kroner. We have aligned single and married filer figures by multiplying the stock income of singles by 2.

Let in Section 2. Importantly, the combined results of this and the previous section suggest that the tax kink is a powerful tool for increasing reported income.

Figure 4: density around tax kinks

7.5 Tables V and VI: prior audits and threat-of-audit letters

Read:

- Table V estimates the effect of prior randomized audits on year-to-year income changes.
- Prior audits increase reported income and reduce future audit adjustments, especially for self-reported income.
- Table VI estimates effects of 50% and 100% audit-threat letters.
- Threat letters increase reported income and reduce underreporting, again mainly where income is self-reported.

Economic message: Enforcement works where taxpayers have reporting discretion. Third-party reporting leaves little room for additional deterrence.

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TABLE IV
TAX EVASION VERSUS TAX AVOIDANCE ELASTICITIES^a

	Baseline		Differences	
	Before Audit Income (Avoidance + Evasion Elasticities)	After Audit Income (Avoidance Elasticity Only)	Difference (Evasion Elasticity Only)	Robustness Check: Difference Using Smaller Sample Around Kink
	1	2	3	4
A. Self-Employment Income (MTR jump from 49% to 62% at 318,700 DKK)				
Fraction bunching (percent)	19.12 (0.90)	12.56 (0.76)	6.56 (1.18)	9.57 (1.86)
Elasticity	0.161 (0.011)	0.085 (0.008)	0.076 (0.014)	0.070 (0.014)
Number of observations	1919	1887	3806	2255
B. Stock Income (MTR jump from 28% to 43% at 88,600 DKK)				
Fraction bunching (percent)	39.30 (2.22)	36.42 (2.11)	2.88 (3.06)	1.80 (3.69)
Elasticity	2.243 (0.213)	1.996 (0.191)	0.247 (0.286)	0.120 (0.259)
Number of observations	486	519	1005	737

^aThis table estimates the effects of marginal tax rates on tax evasion versus tax avoidance using bunching evidence around kink points of the tax schedule where marginal tax rates jump. Panel A focuses on the self-employed and the top rate kink where the marginal tax rates jump from 49% to 62% at 318,700 DKK. Panel B focuses on stock income and the top rate kink for stock income where the marginal tax rates jump from 28% to 43% at 88,600 DKK for married filers and 44,300 DKK for single filers (we have aligned single filers by multiplying by 2 their stock income). As shown in Figure 2, in both cases, there is significant evidence of bunching at the kink both for income before audits and incomes after audits. In each panel, the first row estimates the fraction of tax filers bunching (income within 1500 DKK of the kink) among tax filers with income within 40,000 DKK of the kink. Column 1 is for income before audit while column 2 is for income after audit. Column 3 reports the difference between column 1 and column 2. Column 4 presents a robustness check on the difference when the sample is limited to tax filers within 20,000 DKK (instead of 40,000 DKK) of the kink. In each panel, the second row estimates the (compensated) elasticity of reported income with respect to the net-of-tax rate using bunching evidence (following the method developed in Saez (2010)). Column 1

Table IV: tax evasion versus avoidance elasticities

	Change in Reported Income (Panels A1 and B1) and Probability of Income Increase (Panels A2 and B2) from 2006 to 2007				
	Baseline Audit Adjustment	Total Income	Self-Reported Income	Third-Party Reported Income	IV Effect of Audit Adjustment on Income Change
	1	2	3	4	5
A. Full Sample					
A1. Amounts [difference between the 100% and the 0% audit groups]					
Net income	8491 (827)	2557 (787)	2331 (658)	225 (691)	0.301 (0.098)
Total tax	3295 (257)	1375 (464)			0.417 (0.144)
A2. Probability of audit adjustment and income increase [difference between the 100% and the 0% audit groups]					
Net income	19.09 (0.28)	0.89 (0.48)	2.11 (0.48)	0.24 (0.48)	0.047 (0.025)
Total tax	19.17 (0.28)	0.99 (0.49)			0.052 (0.025)
Number of observations	41,571	41,571	41,571	41,571	41,571
B. Sample Limited to Those Receiving No Threat-of-Audit Letter					
B1. Amounts [difference between the 100% and the 0% audit groups]					
Net income	12,835 (1310)	2904 (1117)	3086 (1008)	-182 (962)	0.226 (0.091)
Total tax	5019 (406)	1732 (677)			0.345 (0.137)

(Continued)

Table V, part 1: effects of randomized prior audits

No. Letter Group	Differences Letter Group vs. No-Letter Group										90% Letter – No Letter	100% Letter – 90% Letter
	Both 0% and 100% Audit Group					0% Audit Group Only					Both 0% and 100% Audit Groups	
	Baseline	Any Adjustment	Upward Adjustment	Downward Adjustment		Any Adjustment	Upward Adjustment	Downward Adjustment			Upward Adjustment	Upward Adjustment
1	2	3	4	5	6	7	8	9	10	11	12	
A. Average Amounts of Individual Upward Adjustments												
Net income	13.37 (31)	1.63 (42)	0.84 (22)	0.84 (24)	10	74 (20)	77 (20)	-3 (45)	115 (15)	92 (152)	23 (28)	58 (28)
Total tax	-322 (24)	67 (32)	50 (18)	17 (26)	17	57 (43)	46 (24)	11 (34)	77 (20)	54 (39)	23 (21)	36 (21)
Number of obs.	3997	24,788	24,788	24,788	14,145	14,145	14,145	10,643	10,643	10,643	24,788	24,788
B. Probability of Upward Adjustments (in percent)												
Net income	13.37 (0.33)	1.63 (0.47)	0.84 (0.28)	0.84 (0.40)	10	74 (0.62)	77 (0.62)	-3 (0.37)	115 (0.55)	92 (0.73)	23 (0.61)	58 (0.33)
Total tax	13.69 (0.35)	1.52 (0.48)	1.57 (0.29)	2.03 (0.40)	1.65 (0.63)	1.65 (0.37)	1.65 (0.54)	0.37 (0.73)	1.02 (0.44)	1.49 (0.61)	1.07 (0.33)	1.07 (0.33)
Number of obs.	3997	24,788	24,788	24,788	14,145	14,145	14,145	10,643	10,643	10,643	24,788	24,788

*The table reports the effects of threat-of-audit letters on individual adjustments to reported income from the time the letter is received in March to the final May file deadline for the tax return filing. Panel A focuses on the average amounts of adjustment. To reduce noise due to extreme observations, all amounts are capped at 10,000 DKK. The cap affects about 1.65 percent of observations for net-income adjustments and 0.75 percent of observations for total tax adjustments (due to net-income adjustments). Panel B focuses on the probability of making an adjustment to net-income or total tax (expressed in percent). Column 1 reports average adjustments (panel A) and probability of adjustment (panel B) among those taxpayers who did not receive the letter. Column 2 reports the difference in average adjustments (panel A) and probability of adjustment (panel B) between the letter and no letter group. Column 3 reports the difference in upward adjustments, while column 4 reports the difference in downward adjustments col. 3 + col. 4 = col. 2. Columns 5, 6, and 7 repeat cols. 2, 3, and 4 but limit the sample to those not audited in the base year (0% audit group). Columns 8, 9, and 10 repeat cols. 2, 3, and 4 but limit the sample to those audited in base year (100% audit group). Column 11 reports the difference in adjustments between the letter group with 50% audit probability and the no-letter group. Column 12 reports the difference in adjustments between the letter group with 100% audit probability and the letter group with 50% audit probability. In each panel, we report effects for net income and for total tax liability. The sample includes only tax filers who did not have any self-employment income in the base year (as tax filers with self-employment income were not part of the letter experiment). Estimates are weighted according to the experiment stratification design. Weights do not reflect population weights. Standard errors are reported in parentheses.

Table VI: threat-of-audit letter effects

8 Short summary

- The paper analyzes a randomized tax audit and audit-threat experiment in Denmark.
- It separates income into third-party-reported and self-reported components.
- Baseline audits show that evasion is close to zero for third-party-reported income but substantial for self-reported income.
- The model explains this by allowing the detection probability to rise sharply once evasion reaches third-party-reported items.
- Bunching around tax kinks shows that marginal tax rates affect evasion, but most taxable-income response is legal avoidance or real behavior.
- Prior audits and threat letters raise later reported income only where taxpayers have discretion.
- Information reporting is the central reason why compliance is high despite low audit probabilities.
- The paper links tax evasion, tax enforcement, third-party reporting, and taxable-income elasticity in one empirical design.

TABLE V—Continued

	Change in Reported Income (Panels A1 and B1) and Probability of Income Increase (Panels A2 and B2) from 2006 to 2007				IV Effect of Audit Adjustment on Income Change
	Baseline Audit Adjustment	Total Income	Self-Reported Income	Third-Party Reported Income	
1	2	3	4	5	
B. Sample Limited to Those Receiving No Threat-of-Audit Letter (Continued)					
B2. Probability of audit adjustment and income increase [difference between the 100% and the 0% audit groups]					
Net income	25.75 (0.39)	0.73 (0.61)	2.12 (0.61)	-0.52 (0.61)	0.028 (0.024)
Total tax	25.93 (0.39)	0.98 (0.61)			0.038 (0.024)
Number of observations	26,180	26,180	26,180	26,180	26,180

*This table reports the effects of prior audits on income changes from 2006 to 2007. Panels A1 and B1 focus on the amounts of income changes while panels A2 and B2 focus on the probability of a (nominal) income increase. In all cases, we report the difference between the 100% audit group and the 0% audit group in the base year. Column 1 reports the difference between the 100% audit group and the 0% audit group in the average amount of audit adjustment in the base year (panels A1 and B1) and the fraction with an income increase for underreporting in base year (panels A2 and B2). Column 2 reports the difference between the 100% audit group and the 0% audit group in the average total reported income. Note that col. 2 = col. 3 + col. 4 for amounts in panels A1 and B1. Column 5 presents the coefficient of an IV regression of income change (panels A1 and B1) and dummy for an income increase (panels A2 and B2) on the baseline audit adjustment for underreporting used as instrument the 100% audit group dummy. Effectively, we have col. 5 = col. 2/col. 1. This coefficient in panels A1 and B1 can be interpreted as the causal effect of an additional dollar of audit adjustment on reported income. We have col. 5 = col. 2/col. 1. This coefficient in panels A1 and B1 can be interpreted as the causal effect of an additional dollar of audit adjustment on reported income. Weights do not reflect population weights. Standard errors are reported in parentheses. For panels A1 and B1, all the amounts are in Danish kroner (U.S. \$1 = 5.2 DKK as of 1/2010). Income changes are trimmed at -200,000 DKK and 200,000 DKK. That is, income changes are defined as min(200,000, max(income in 2007 – income in 2006, -200,000)). This is done to avoid extreme outcomes, which make estimates very imprecise. Less than 2% of observations are trimmed on average.

Table V, part 2: prior audits continued

	A. Audit Randomization				B. Letter Randomization				C. Within Letter Randomization			
	0% Audit Group	100% Audit Group	Difference Standard Error		No-Letter Group	100% Letter Group	Difference Standard Error		30% Letter Group	100% Letter Group	Difference Standard Error	
	1	2	3	4	5	6	7	8	9	10	11	12
Net income	248,290	263,485	-1724	(6047)	229,036	244,477	-4541	(3423)	243,878	243,078	800	(4422)
Total tax	100,960	100,460	-508	(3010)	82,443	84,230	1786	(1588)	80,022	84,438	4415	(2073)
Personal income	214,434	217,426	1007	(2351)	217,022	219,748	2725	(2804)	219,374	240,123	20,749	(1709)
Capital income	-13,127	-13,895	-768	(1015)	-16,514	-15,485	1029	(554)	-15,613	-15,308	305	(626)
Deductions	-11,839	-11,976	-138	(160)	-8333	-8304	29	(140)	-8268	-8341	-73	(193)
Stock income	18,141	15,800	-2341	(4928)	17,671	8220	8459	(1777)	7857	8594	737	(2243)
Self-employment	55,616	54,960	-656	(2869)	430	299	-131	(299)	577	79	-497	(268)
% with net income	99.55	99.52	-0.03	(0.07)	98.73	98.64	-0.09	(0.15)	98.52	98.76	0.24	(0.19)
% with total tax	96.71	96.61	-0.11	(0.17)	96.64	96.26	-0.38	(0.25)	96.26	96.25	-0.02	(0.31)
% with personal income	94.98	94.85	-0.13	(0.21)	97.29	97.11	-0.18	(0.22)	96.99	97.23	0.23	(0.27)
% with capital income	95.67	95.40	-0.27	(0.20)	97.62	96.90	-0.72	(0.23)	96.77	97.03	0.26	(0.28)
% with deductions	71.69	71.76	0.07	(0.44)	64.18	64.49	0.31	(0.65)	64.76	64.19	-0.60	(0.77)
% with stock income	40.30	40.23	-0.07	(0.47)	44.07	43.63	-0.44	(0.67)	43.59	43.68	0.09	(0.80)
% with self-employment	40.18	40.27	0.19	(0.47)	0.78	0.79	0.01	(0.12)	0.77	0.82	0.05	(0.14)
Female (%)	39.95	39.59	-0.33	(0.47)	49.80	50.10	0.30	(0.67)	49.83	50.38	0.55	(0.81)
Married (%)	58.46	58.13	-0.32	(0.48)	54.54	53.22	-1.32	(0.67)	53.79	52.65	-1.13	(0.86)
Church membership (%)	82.83	83.71	0.12	(0.34)	86.82	86.86	0.04	(0.46)	87.06	86.66	-0.40	(0.54)
Copenhagen (%)	3.14	3.13	-0.01	(0.17)	3.17	3.33	0.16	(0.24)	3.32	3.34	0.02	(0.29)
Age	49.28	49.43	0.14	(0.16)	49.09	49.90	0.81	(0.25)	49.01	48.80	-0.21	(0.30)
% filing in 2007	97.08	96.94	-0.14	(0.16)	100.00	100.00	0.00	(0.00)	100.00	100.00	0.00	(0.00)
Number of observation	22,148	16,630	42,778		9397	15,391	24,788		7706	7685	15,391	

*This table presents randomization checks for the audit experiment (part A, columns 1-4) and the letter experiment (part B, columns 5-8 and part C, columns 9-12). Part A compares baseline reported income in 2006 (before the audit experiment took place). Columns 1 and 2 present the baseline averages for the treatment group and control group, respectively. Column 3 presents the difference between the treatment group and the control group. The standard error of the difference is presented in column 4. Parts B and C compare preperiod tax returns for 2007 incomes before the letters are sent. The columns in parts B and C are constructed as in part A. In part B, the sample is restricted to tax filers not registered as self-employed in the base year as the letter experiment could not be carried out for self-employed. In part C, the sample is further restricted to tax filers who received either the 50% threat-of-audit letter or the 100% threat-of-audit letter. Estimates are weighted according to the experiment stratification design. Weights do not reflect population weights. All the amounts are in Danish kroner (U.S. \$1 = 5.2 DKK as of 1/2010).

9 Conclusion

9.1 Core conclusion

Kleven, Knudsen, Kreiner, Pedersen, and Saez show that the key determinant of tax compliance is not simply taxpayer morality or the statutory audit rate. It is the information environment. When income is third-party reported, evasion is nearly impossible. When income is self-reported, evasion is common and responsive to enforcement.

9.2 Economic intuition

The standard evasion model predicts much more noncompliance than we observe because it treats detection as a low exogenous probability. In modern tax systems, third-party reports make detection highly likely for many income items. Taxpayers are therefore unable to cheat on most income, and enforcement resources matter most for self-reported margins.

9.3 Takeaway

The paper's central message is that tax capacity depends on information systems. Audits and penalties matter, but the largest compliance gains come from reducing the set of items over which taxpayers have private information.